Solutions to Judson's *Abstract Algebra*
Chapter 14: Group Actions, Exercises 1–13

Ali

Preliminaries: conventions and tools

Throughout, G is a group acting on a set X on the left, so $ex = x$ and $(g_1g_2)x = g_1(g_2x)$.

- $x \sim_G y$ (G-equivalent) means $gx = y$ for some $g \in G$. The equivalence classes are the *orbits* $\mathcal{O}_x = \{gx : g \in G\}$.
- $X_g = \{x \in X : gx = x\} \subseteq X$ is the *fixed point set* of g.
- $G_x = \{g \in G : gx = x\} \leq G$ is the *stabilizer* (isotropy subgroup) of x.

For a permutation group $G \leq S_X$ the action is $\sigma \cdot x = \sigma(x)$.

Theorem 1 (Orbit–Stabilizer, Judson Thm. 14.11)**.** *If G and X are finite and $x \in X$, then $|\mathcal{O}_x| = [G : G_x] = |G|/|G_x|$; equivalently $|G| = |\mathcal{O}_x| \cdot |G_x|$.*

Theorem 2 (Burnside, Judson Thm. 14.19)**.** *If G and X are finite, the number of orbits of G on X is*

$$k \;=\; \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Counting colorings

Let S be a finite set (vertices, edges, or faces of a solid) on which G acts, and let C be a finite set of colors. A coloring is a function $f\colon S \to C$, and G acts on colorings by $(g \cdot f)(s) = f(g^{-1}s)$. Two colorings are regarded as the same exactly when they lie in the same orbit, so the number of distinct colorings is the number of orbits.

A coloring f is fixed by g iff $f(g^{-1}s) = f(s)$ for all s, i.e. iff f is constant on every cycle of g acting on S. Hence

$$|\mathrm{Fix}(g)| = |C|^{c(g)}, \qquad c(g) = \text{number of cycles of } g \text{ on } S \text{ (including 1-cycles).} \tag{1}$$

The rotation group of the cube

Let G be the group of rotations of a cube. Since G acts transitively on the 6 faces and the stabilizer of a face consists of the 4 rotations about the axis through that face, Theorem 1 gives $|G| = 6 \cdot 4 = 24$. The 24 rotations fall into five types:

Type	Axis	Number
identity	—	1
$\pm 90^\circ$	through centers of opposite faces (3 axes)	6
180°	through centers of opposite faces (3 axes)	3
$\pm 120^\circ$	through opposite vertices (4 axes)	8
180°	through midpoints of opposite edges (6 axes)	6

$(1+6+3+8+6=24.)$ Their cycle structures on faces, vertices and edges are (writing a^b for b cycles of length a):

Type	Number	Faces (6)	Vertices (8)	Edges (12)
identity	1	1^6	1^8	1^{12}
$\pm 90°$ face	6	$1^2\,4^1$	4^2	4^3
$180°$ face	3	$1^2\,2^2$	2^4	2^6
$\pm 120°$ vertex	8	3^2	$1^2\,3^2$	3^4
$180°$ edge	6	2^3	2^4	$1^2\,2^5$

Number of cycles $c(g)$: faces $6,3,4,2,3$; vertices $8,2,4,4,4$; edges $12,3,6,4,7$ (in the order of the rows). For instance, a $180°$ rotation about an axis through the midpoints of opposite edges e, e' fixes e and e' setwise (reversing each) and swaps the remaining ten edges in five pairs, giving $1^2 2^5$.

Exercises

Exercise 1. *Examples 14.1–14.5 each describe an action of a group G on a set X. For each example, compute the G-equivalence classes (orbits).*

Solution. Recall that the G-equivalence class of x is its orbit $\mathcal{O}_x = \{gx : g \in G\}$.

Example 14.1 ($G = GL_2(\mathbb{R})$, $X = \mathbb{R}^2$, $g \cdot v = gv$). We claim there are exactly two orbits: $\{0\}$ and $\mathbb{R}^2 \setminus \{0\}$. First, $A0 = 0$ for every $A \in G$, so $\mathcal{O}_0 = \{0\}$. Now let $v, w \in \mathbb{R}^2$ be nonzero. Choose $v', w' \in \mathbb{R}^2$ so that $\{v, v'\}$ and $\{w, w'\}$ are bases of $\mathbb{R}^2$, and let $B_v = [\,v\ v'\,]$, $B_w = [\,w\ w'\,]$ (matrices with these columns). Both are invertible, so $A = B_w B_v^{-1} \in GL_2(\mathbb{R})$, and $Av = B_w B_v^{-1} B_v \mathbf{e}_1 = B_w \mathbf{e}_1 = w$. Thus every nonzero vector is G-equivalent to every other, so $\mathcal{O}_v = \mathbb{R}^2 \setminus \{0\}$ for $v \neq 0$.

Example 14.2 ($G = D_4 = \{(1), (13), (24), (1432), (1234), (12)(34), (14)(23), (13)(24)\}$, $X = \{1,2,3,4\}$). There is a single orbit X: the element $(1234) \in G$ sends $1 \mapsto 2$, $(1234)^2 = (13)(24)$ sends $1 \mapsto 3$, and $(1234)^3 = (1432)$ sends $1 \mapsto 4$. So $\mathcal{O}_1 = \{1,2,3,4\}$, and since orbits partition X, this is the only G-equivalence class.

Example 14.3 ($X = G$, $g \cdot x = gx$). Given $x, y \in G$, take $g = yx^{-1}$; then $gx = y$. So there is one orbit, $\mathcal{O}_x = G$ for every x.

Variant: $H \leq G$ acting on G by left multiplication. Here $\mathcal{O}_x = \{hx : h \in H\} = Hx$, the *right coset* of H containing x. So the H-equivalence classes are the right cosets of H in G, and there are $[G : H]$ of them.

Example 14.4 ($H \leq G$ acting on $X = G$ by conjugation, $h \cdot g = hgh^{-1}$). The orbit of g is $\mathcal{O}_g = \{hgh^{-1} : h \in H\}$, the *$H$-conjugacy class* of g, and these classes are the G-equivalence classes. Its size is $|\mathcal{O}_g| = [H : H \cap C_G(g)]$ by Theorem 1, since the stabilizer of g is $H \cap C_G(g)$. In particular $\mathcal{O}_g = \{g\}$ iff g commutes with every element of H. When $H = G$ the orbits are the usual conjugacy classes of G.

Illustration: $G = S_3$, $H = A_3 = \{(1), (123), (132)\}$. Since A_3 is abelian, (123) and (132) are each fixed by H. For a transposition, $(123)(12)(123)^{-1} = (\,(123)(1)\ (123)(2)\,) = (23)$ and $(132)(12)(132)^{-1} = (13)$. So the orbits are

$$\{(1)\}, \quad \{(123)\}, \quad \{(132)\}, \quad \{(12), (13), (23)\}.$$

Example 14.5 ($H \leq G$, $X = \mathcal{L}_H$ the set of left cosets, $g \cdot xH = gxH$). Given $xH, yH \in \mathcal{L}_H$, let $g = yx^{-1}$; then $g \cdot xH = yH$. So there is exactly one orbit, namely all of $\mathcal{L}_H$. ■

Exercise 2. *Compute all X_g and all G_x for each of the following permutation groups. (a) $X = \{1,2,3\}$, $G = S_3 = \{(1),(12),(13),(23),(123),(132)\}$. (b) $X = \{1,2,3,4,5,6\}$, $G = \{(1),(12),(345),(354),(12)(345),(12)(354)\}$.*

Solution. In both parts the action is $\sigma \cdot x = \sigma(x)$. A point x is fixed by σ exactly when x does not appear in a nontrivial cycle of σ.

(a) Fixed point sets:

g	(1)	(12)	(13)	(23)	(123)	(132)
X_g	$\{1,2,3\}$	$\{3\}$	$\{2\}$	$\{1\}$	$\emptyset$	$\emptyset$

Stabilizers (collect the g with $x \in X_g$):

$$G_1 = \{(1),(23)\}, \qquad G_2 = \{(1),(13)\}, \qquad G_3 = \{(1),(12)\}.$$

(b) Fixed point sets:

g	(1)	(12)	(345)	(354)	$(12)(345)$	$(12)(354)$
X_g	X	$\{3,4,5,6\}$	$\{1,2,6\}$	$\{1,2,6\}$	$\{6\}$	$\{6\}$

Stabilizers:

$$\begin{aligned} G_1 = G_2 &= \{(1),(345),(354)\}, \\ G_3 = G_4 = G_5 &= \{(1),(12)\}, \\ G_6 &= G. \end{aligned}$$

■

Exercise 3. *Compute the G-equivalence classes of X for each of the G-sets in Exercise 2. For each $x \in X$ verify that $|G| = |\mathcal{O}_x| \cdot |G_x|$.*

Solution. **(a)** $G = S_3$ contains (12), (13) and (23), so $1 \sim 2$, $1 \sim 3$, and there is one orbit $\mathcal{O}_1 = \mathcal{O}_2 = \mathcal{O}_3 = \{1,2,3\}$. Here $|G| = 6$ and

$$|\mathcal{O}_x| \cdot |G_x| = 3 \cdot 2 = 6 \quad \text{for } x = 1,2,3,$$

using $|G_1| = |G_2| = |G_3| = 2$ from Exercise 2(a).

(b) The elements of G act as follows: (12) swaps $1,2$; (345) and (354) permute $\{3,4,5\}$ cyclically; and 6 is fixed by everything. So the orbits are

$$\mathcal{O}_1 = \mathcal{O}_2 = \{1,2\}, \qquad \mathcal{O}_3 = \mathcal{O}_4 = \mathcal{O}_5 = \{3,4,5\}, \qquad \mathcal{O}_6 = \{6\}.$$

Check of $|G| = 6 = |\mathcal{O}_x| \cdot |G_x|$:

x	$\lvert\mathcal{O}_x\rvert$	$\lvert G_x\rvert$	$\lvert\mathcal{O}_x\rvert \cdot \lvert G_x\rvert$
$1,2$	2	3	6
$3,4,5$	3	2	6
6	1	6	6

■

Exercise 4. *Let G be the additive group of real numbers. Let the action of $\theta \in G$ on the real plane $\mathbb{R}^2$ be given by rotating the plane counterclockwise about the origin through θ radians. Let $P \neq 0$ be a point of the plane. (a) Show that $\mathbb{R}^2$ is a G-set. (b) Describe geometrically the orbit containing P. (c) Find the group G_P.*

Solution. **(a)** Let $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and define $\theta \cdot Q = R_\theta Q$. We check the two axioms for the group $(\mathbb{R}, +)$, whose identity is 0.

- $R_0 = I$, so $0 \cdot Q = Q$.
- By the angle-addition formulas,

$$R_\theta R_\varphi = \begin{pmatrix} \cos\theta\cos\varphi - \sin\theta\sin\varphi & -(\cos\theta\sin\varphi + \sin\theta\cos\varphi) \\ \sin\theta\cos\varphi + \cos\theta\sin\varphi & \cos\theta\cos\varphi - \sin\theta\sin\varphi \end{pmatrix} = R_{\theta+\varphi},$$

  hence $(\theta + \varphi) \cdot Q = R_{\theta+\varphi}Q = R_\theta(R_\varphi Q) = \theta \cdot (\varphi \cdot Q)$.

So $\mathbb{R}^2$ is a G-set.

(b) Write $P = r(\cos\alpha, \sin\alpha)$ with $r = |P| > 0$. Then $R_\theta P = r(\cos(\alpha+\theta), \sin(\alpha+\theta))$. As θ ranges over $\mathbb{R}$, $\alpha + \theta$ ranges over all angles, so

$$\mathcal{O}_P = \{\, Q \in \mathbb{R}^2 : |Q| = |P| \,\},$$

the *circle centered at the origin passing through P* (radius $|P|$). Indeed, rotations preserve distance from the origin, and any $Q = r(\cos\beta, \sin\beta)$ on this circle equals $R_{\beta-\alpha}P$.

(c) $\theta \in G_P$ iff $\cos(\alpha+\theta) = \cos\alpha$ and $\sin(\alpha+\theta) = \sin\alpha$ (as $r > 0$), iff $\theta \in 2\pi\mathbb{Z}$. Hence

$$G_P = 2\pi\mathbb{Z} = \{2\pi n : n \in \mathbb{Z}\}.$$

This is the same subgroup for every $P \neq 0$. ■

Exercise 5. *Let $G = A_4$ act on itself by conjugation, $(g, h) \mapsto ghg^{-1}$. (a) Determine the conjugacy classes (orbits) of each element of G. (b) Determine all of the isotropy subgroups for each element of G.*

Solution. A_4 has 12 elements: the identity; the three double transpositions $(12)(34), (13)(24), (14)(23)$ (together with (1) they form the Klein four-group V); and the eight 3-cycles. We use the rule $\sigma(a\,b\,c)\sigma^{-1} = (\sigma(a)\ \sigma(b)\ \sigma(c))$, and likewise for products of cycles. The isotropy subgroup of h is its centralizer $C(h) = \{g \in A_4 : gh = hg\}$.

Identity. $\mathcal{O}_{(1)} = \{(1)\}$ and $C((1)) = A_4$.

Double transpositions. Conjugating $(12)(34)$ by (123) gives $(23)(14) = (14)(23)$, and by (132) gives $(31)(24) = (13)(24)$. So the orbit contains all three double transpositions, and

$$\mathcal{O}_{(12)(34)} = \{(12)(34), (13)(24), (14)(23)\}.$$

By Theorem 1, $|C((12)(34))| = 12/3 = 4$. Since V is abelian of order 4 and contains $(12)(34)$, we get $C((12)(34)) = V$. The same argument applies to the other two, so each double transposition has isotropy subgroup $V = \{(1), (12)(34), (13)(24), (14)(23)\}$.

3-cycles. Conjugating (123) by the elements of V:

$$(12)(34)\,(123)\,(12)(34)^{-1} = (2\,1\,4) = (142),$$
$$(13)(24)\,(123)\,(13)(24)^{-1} = (3\,4\,1) = (134),$$
$$(14)(23)\,(123)\,(14)(23)^{-1} = (4\,3\,2) = (243).$$

So the orbit of (123) has at least 4 elements. Its centralizer contains $\langle(123)\rangle$ of order 3, so $|\mathcal{O}_{(123)}| = 12/|C((123))| \le 12/3 = 4$. Therefore

$$\mathcal{O}_{(123)} = \{(123), (142), (134), (243)\}, \qquad C((123)) = \langle(123)\rangle.$$

Taking inverses, $\mathcal{O}_{(132)} = \{(132), (124), (143), (234)\}$ (these are the inverses of the previous four, and by the same argument it has exactly 4 elements). Note that (123) and (132) are *not* conjugate in A_4.

(a) Conjugacy classes of A_4:

$$\{(1)\}, \quad \{(12)(34), (13)(24), (14)(23)\}, \quad \{(123), (142), (134), (243)\}, \quad \{(132), (124), (143), (234)\}.$$

(Class equation: $12 = 1 + 3 + 4 + 4$.)

(b) Isotropy subgroups. For a 3-cycle g, $C(g)$ has order 3 and contains $\langle g\rangle$, so $C(g) = \langle g\rangle$:

Element h	Isotropy subgroup $G_h = C(h)$
(1)	A_4
(12)(34), (13)(24), (14)(23)	$V = \{(1), (12)(34), (13)(24), (14)(23)\}$
(123), (132)	$\{(1), (123), (132)\}$
(134), (143)	$\{(1), (134), (143)\}$
(142), (124)	$\{(1), (142), (124)\}$
(243), (234)	$\{(1), (243), (234)\}$

■

Exercise 6. *Find the conjugacy classes and the class equation for each of the following groups: (a) S_4, (b) D_5, (c) $\mathbb{Z}_9$, (d) Q_8.*

Solution. **(a) S_4.** Two permutations in S_n are conjugate iff they have the same cycle type. The classes of S_4 ($|S_4| = 24$):

Cycle type	Size	Count
1^4 (identity)	1	1
$2\,1^2$ (transpositions)	6	$\binom{4}{2} = 6$
2^2 (double transpositions)	3	$\frac{1}{2}\binom{4}{2} = 3$
$3\,1$ (3-cycles)	8	$\binom{4}{3} \cdot 2 = 8$
4 (4-cycles)	6	$3! = 6$

Class equation: $24 = 1 + 3 + 6 + 6 + 8$.

(b) D_5 (order 10). Write $D_5 = \{r^i s^j : 0 \le i \le 4,\ j = 0, 1\}$ with $r^5 = s^2 = \mathrm{e}$, $srs^{-1} = r^{-1}$.

- e is central: class $\{e\}$.
- Rotations: $sr^k s^{-1} = r^{-k}$, and conjugating by powers of r does nothing to r^k (they commute). So $\{r, r^4\}$ and $\{r^2, r^3\}$ are classes.
- Reflections: $r\,(r^j s)\,r^{-1} = r^{j+1}sr^{-1} = r^{j+1}r\,s = r^{j+2}s$. Since 2 generates $\mathbb{Z}_5$, the conjugates of s by powers of r include all five reflections s, rs, r^2s, r^3s, r^4s. Hence the five reflections form one class.

Class equation: $10 = 1 + 2 + 2 + 5$.

(c) $\mathbb{Z}_9$**.** The group is abelian, so $gxg^{-1} = x$ for all g, and every element is its own class: nine classes of size 1. Class equation: $9 = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$.

(d) $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = -1$, $ij = k$. The elements ± 1 are central, giving classes $\{1\}$ and $\{-1\}$. Since $jij^{-1} = ji(-j) = -(ji)j = -(-k)j = kj = -i$, we get $\{i, -i\}$ as a class, and similarly $\{j, -j\}$ and $\{k, -k\}$ (each has centralizer $\langle\cdot\rangle$ of order 4, so class size $8/4 = 2$). Classes:

$$\{1\},\ \{-1\},\ \{i, -i\},\ \{j, -j\},\ \{k, -k\}.$$

Class equation: $8 = 1 + 1 + 2 + 2 + 2$. ■

Exercise 7. *Write the class equation for S_5 and for A_5.*

Solution. S_5 ($|S_5| = 120$). Conjugacy classes correspond to cycle types:

Cycle type	Size	Count
1^5	1	1
$2\,1^3$	10	$\binom{5}{2}$
$2^2\,1$	15	$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$
$3\,1^2$	20	$\binom{5}{3} \cdot 2$
$3\,2$	20	$\binom{5}{3} \cdot 2 \cdot 1$
$4\,1$	30	$\binom{5}{4} \cdot 3!$
5	24	$4!$

Class equation: $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$.

A_5 ($|A_5| = 60$). We use the following fact.

Lemma 3. *Let $g \in A_n$ and let K be its conjugacy class in S_n. If $C_{S_n}(g) \not\subseteq A_n$, then K is a single conjugacy class of A_n. If $C_{S_n}(g) \subseteq A_n$, then K splits into two A_n-classes of size $|K|/2$ each.*

Proof. The A_n-class of g has size $|A_n|/|C_{A_n}(g)|$ with $C_{A_n}(g) = C_{S_n}(g) \cap A_n$, and $|K| = n!/|C_{S_n}(g)|$. If $C_{S_n}(g) \not\subseteq A_n$, then $C_{S_n}(g) \cap A_n$ has index 2 in $C_{S_n}(g)$, so the A_n-class has size $\frac{n!/2}{|C_{S_n}(g)|/2} = |K|$; being contained in K, it equals K. If $C_{S_n}(g) \subseteq A_n$, the A_n-class has size $\frac{n!/2}{|C_{S_n}(g)|} = |K|/2$, so K splits into two classes (the two A_n-classes are swapped by conjugation by any odd permutation). □

The even cycle types in S_5 are 1^5, $2^2 1$, $3\,1^2$, and 5.

- *Identity:* class of size 1.
- $(12)(34)$*:* $|C_{S_5}| = 120/15 = 8$, and this centralizer contains the odd permutation (12). No splitting: one class of size 15.
- (123)*:* $|C_{S_5}| = 120/20 = 6$, and $C_{S_5}((123)) = \langle(123)\rangle \times \langle(45)\rangle$ contains the odd permutation (45). No splitting: one class of size 20.

- (12345)*:* $|C_{S_5}| = 120/24 = 5$, so $C_{S_5}((12345)) = \langle(12345)\rangle \subseteq A_5$. The class splits into two A_5-classes of size 12.

Class equation: $60 = 1 + 12 + 12 + 15 + 20$. ■

Exercise 8. *If a square remains fixed in the plane, how many different ways can the corners of the square be colored if three colors are used?*

Solution. Label the corners $1, 2, 3, 4$ in order around the square, so $S = \{1, 2, 3, 4\}$ and C has $|C| = 3$ colors. Because the square stays in the plane, the only motions that can identify two colorings are the *rotations* of the square in the plane (flipping the square over would take it out of the plane). These form the cyclic group

$$G = \{(1),\ (1234),\ (13)(24),\ (1432)\} \cong \mathbb{Z}_4.$$

By (1), a rotation with c cycles fixes 3^c colorings:

g	(1)	(1234)	(13)(24)	(1432)
$c(g)$	4	1	2	1
$\lvert\mathrm{Fix}(g)\rvert$	81	3	9	3

By Burnside's theorem the number of distinct colorings is

$$\frac{1}{4}\,(81 + 3 + 9 + 3) = \frac{96}{4} = \boxed{24}.$$

Remark. This reads "fixed in the plane" as ruling out flips. If one also allowed the reflections of D_4 (turning the square over), the two diagonal reflections each have 3 cycles (27 fixed colorings) and the two edge-midpoint reflections each have 2 cycles (9 fixed colorings), giving $\frac{1}{8}(81 + 3 + 9 + 3 + 2 \cdot 27 + 2 \cdot 9) = \frac{168}{8} = 21$. Check which reading your edition's hints use.

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Exercise 9. *How many ways can the vertices of an equilateral triangle be colored using three different colors?*

Solution. Here the triangle is a free object, so it may be rotated or turned over; the symmetry group is $D_3 \cong S_3$ acting on the vertex set $S = \{1, 2, 3\}$. We interpret "three colors" as three available colors ($|C| = 3$) that need not all be used. By (1):

elements	number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert$
identity	1	3	27
rotations $(123), (132)$	2	1	3
reflections $(12), (13), (23)$	3	2	9

By Burnside's theorem the number of distinct colorings is

$$\frac{1}{6}\,(27 + 2 \cdot 3 + 3 \cdot 9) = \frac{60}{6} = \boxed{10}.$$

Remark. If only rotations were allowed, the count would be $\frac{1}{3}(27 + 3 + 3) = 11$. If instead one wants all three colors used on the three vertices, there is exactly 1 way up to symmetry under D_3 (and 2 up to rotation only).

■

Exercise 10. *Find the number of ways a six-sided die can be constructed if each side is marked differently with* $1, \ldots, 6$ *dots.*

Solution. Let G be the rotation group of the cube ($|G| = 24$). A die is a labeling f of the six faces by the numbers $1, \ldots, 6$ with each number used exactly once, so there are $6! = 720$ labelings. Two labelings give the same die iff one is obtained from the other by a rotation, so we count the orbits of G on the set X of these 720 labelings.

Only the identity fixes a labeling. If $g \cdot f = f$, then $f(g^{-1}F) = f(F)$ for every face F. Since f is injective, $g^{-1}F = F$ for every face, i.e. g fixes all six faces. The only rotation doing so is the identity (it fixes the three coordinate axes pointwise, hence is the identity map). Thus $|\mathrm{Fix}(\mathrm{e})| = 720$ and $|\mathrm{Fix}(g)| = 0$ for $g \neq \mathrm{e}$.

By Burnside's theorem the number of dice is

$$\frac{1}{24} \cdot 720 = \boxed{30}.$$

■

Exercise 11. *Up to a rotation, how many ways can the faces of a cube be colored with three different colors?*

Solution. Let G be the rotation group of the cube acting on the six faces, with $|C| = 3$ colors. Using the face cycle counts from the table in the Preliminaries and (1):

Type	Number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert = 3^{c(g)}$
identity	1	6	729
$\pm 90°$ face	6	3	27
$180°$ face	3	4	81
$\pm 120°$ vertex	8	2	9
$180°$ edge	6	3	27

Burnside's theorem gives

$$\frac{1}{24}\big(1 \cdot 729 + 6 \cdot 27 + 3 \cdot 81 + 8 \cdot 9 + 6 \cdot 27\big) = \frac{729 + 162 + 243 + 72 + 162}{24} = \frac{1368}{24} = \boxed{57}.$$

■

Exercise 12. *Consider* 12 *straight wires of equal lengths with their ends soldered together to form the edges of a cube. Either silver or copper wire can be used for each edge. How many different ways can the cube be constructed?*

Solution. The rotation group G of the cube acts on the 12 edges; we color each edge with one of $|C| = 2$ colors (silver or copper), and two constructions are the same iff they differ by a rotation. Using the edge cycle counts from the Preliminaries:

Type	Number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert = 2^{c(g)}$
identity	1	12	4096
$\pm 90°$ face	6	3	8
$180°$ face	3	6	64
$\pm 120°$ vertex	8	4	16
$180°$ edge	6	7	128

Hence the number of constructions is

$$\frac{1}{24}\big(4096 + 6\cdot 8 + 3\cdot 64 + 8\cdot 16 + 6\cdot 128\big) = \frac{4096+48+192+128+768}{24} = \frac{5232}{24} = \boxed{218}.$$

■

Exercise 13. *Suppose that we color each of the eight corners of a cube. Using three different colors, how many ways can the corners be colored up to a rotation of the cube?*

Solution. Now G acts on the 8 vertices and $|C| = 3$. Using the vertex cycle counts from the Preliminaries:

Type	Number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert = 3^{c(g)}$
identity	1	8	6561
$\pm 90°$ face	6	2	9
$180°$ face	3	4	81
$\pm 120°$ vertex	8	4	81
$180°$ edge	6	4	81

By Burnside's theorem the number of colorings is

$$\frac{1}{24}\big(6561 + 6\cdot 9 + 3\cdot 81 + 8\cdot 81 + 6\cdot 81\big) = \frac{6561+54+243+648+486}{24} = \frac{7992}{24} = \boxed{333}.$$

■

Exercise 14. *Each of the faces of a regular tetrahedron can be painted either red or white. Up to a rotation, how many different ways can the tetrahedron be painted?*

Solution. Let G be the rotation group of the regular tetrahedron. It acts transitively on the 4 faces, and the stabilizer of a face consists of the 3 rotations about the axis through that face's center, so $|G| = 4\cdot 3 = 12$ by Theorem 1. The rotations are:

- the identity (1 element);
- $\pm 120°$ rotations about an axis through a vertex and the center of the opposite face (4 axes, 8 elements): the opposite face is fixed and the other three faces are permuted in a 3-cycle, so the cycle type on faces is $1\cdot 3$, with $c = 2$;
- $180°$ rotations about an axis through the midpoints of two opposite edges (3 axes, 3 elements): with vertices a, b, c, d and axis through the midpoints of ab and cd, the rotation swaps the faces $abc \leftrightarrow abd$ and $acd \leftrightarrow bcd$, so the cycle type is 2^2, with $c = 2$.

$(1 + 8 + 3 = 12.)$ With $|C| = 2$ colors, (1) gives

Type	Number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert = 2^{c(g)}$
identity	1	4	16
$\pm 120°$	8	2	4
$180°$	3	2	4

By Burnside's theorem the number of colorings is

$$\frac{1}{12}\big(16 + 8\cdot 4 + 3\cdot 4\big) = \frac{60}{12} = \boxed{5}.$$

(Explicitly: 0, 1, 2, 3, or 4 red faces gives one orbit each.) ■

Exercise 15. *Suppose that the vertices of a regular hexagon are to be colored either red or white. How many ways can this be done up to a symmetry of the hexagon?*

Solution. The symmetry group of the regular hexagon is D_6, of order 12, acting on the vertex set $S = \{1, \dots, 6\}$. Its elements and their cycle counts on the vertices are:

Type	Number	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert = 2^{c(g)}$
identity	1	6	64
rotations by $\pm 60°$ (one 6-cycle)	2	1	2
rotations by $\pm 120°$ (two 3-cycles)	2	2	4
rotation by $180°$ (three 2-cycles)	1	3	8
reflections through opposite vertices (2 fixed vertices, two 2-cycles)	3	4	16
reflections through midpoints of opposite edges (three 2-cycles)	3	3	8

$(1 + 2 + 2 + 1 + 3 + 3 = 12.)$ By Burnside's theorem the number of colorings is

$$\frac{1}{12}\big(64 + 2 \cdot 2 + 2 \cdot 4 + 8 + 3 \cdot 16 + 3 \cdot 8\big) = \frac{64 + 4 + 8 + 8 + 48 + 24}{12} = \frac{156}{12} = \boxed{13}.$$

■

Exercise 16. *A molecule of benzene is made up of six carbon atoms and six hydrogen atoms, linked together in a hexagonal shape. (a) How many different compounds can be formed by replacing one or more of the hydrogen atoms with a chlorine atom? (b) Find the number of different chemical compounds that can be formed by replacing three of the six hydrogen atoms in a benzene ring with a* CH_3 *radical.*

Solution. The six hydrogen positions sit at the vertices of a regular hexagon, and the molecule may be rotated and turned over in space, so two substitution patterns give the same compound iff they differ by an element of D_6 (order 12), whose cycle structure on the six positions is tabulated in Exercise 15.

(a) A substitution pattern is a coloring of the six positions with two "colors" (H or Cl). By Exercise 15 there are 13 patterns up to symmetry. One of them is the all-H pattern, which is benzene itself and involves no replacement. Hence the number of compounds with one or more chlorine atoms is

$$13 - 1 = \boxed{12}.$$

(b) Now X is the set of 3-element subsets of the six positions (the positions carrying CH_3); $|X| = \binom{6}{3} = 20$. A subset is fixed by g iff it is a union of cycles of g. For each type of element:

- identity: all 20 subsets are fixed.
- rotations by $\pm 60°$ (6-cycle): no union of cycles has size 3, so 0.
- rotations by $\pm 120°$ (two 3-cycles): the two 3-cycles themselves, so 2 each.
- rotation by $180°$ (three 2-cycles): sizes are even, so 0.
- reflections through vertices (two 1-cycles, two 2-cycles): size $3 = 1 + 2$, choose one fixed point and one 2-cycle: $2 \cdot 2 = 4$ each.
- reflections through edge midpoints (three 2-cycles): sizes are even, so 0.

By Burnside's theorem the number of compounds is

$$\frac{1}{12}\big(20 + 2\cdot 0 + 2\cdot 2 + 0 + 3\cdot 4 + 3\cdot 0\big) = \frac{36}{12} = \boxed{3}.$$

These are the three substitution patterns with the CH_3 groups in positions $\{1,2,3\}$, $\{1,2,4\}$, and $\{1,3,5\}$. ■

Exercise 17. *How many equivalence classes of switching functions are there if the input variables x_1, x_2, x_3 can be permuted by any permutation in S_3? What if the input variables x_1, x_2, x_3, x_4 can be permuted by any permutation in S_4?*

Solution. A switching function of n variables is a function $f\colon V \to \{0,1\}$ where $V = \{0,1\}^n$ is the set of 2^n input vectors. So the switching functions are exactly the 2-colorings of V. A permutation $\sigma \in S_n$ acts on V by permuting coordinates, and hence on functions by $(\sigma \cdot f)(v) = f(\sigma^{-1}v)$. Two functions are equivalent iff they are in the same orbit. By (1),

$$|\mathrm{Fix}(\sigma)| = 2^{c_V(\sigma)},$$

where $c_V(\sigma)$ is the number of cycles of σ acting on V.

Computing $c_V(\sigma)$. The cycles of σ on V are the orbits of $\langle\sigma\rangle$ on V, so by Burnside's theorem applied to $\langle\sigma\rangle$,

$$c_V(\sigma) = \frac{1}{\mathrm{ord}(\sigma)} \sum_{k=0}^{\mathrm{ord}(\sigma)-1} |V_{\sigma^k}|,$$

where $|V_\tau| = 2^{c(\tau)}$ (a vector is fixed by τ iff it is constant on the cycles of τ acting on the coordinates, so $c(\tau)$ is the number of cycles of τ on $\{1,\dots,n\}$).

$n = 3$ ($|V| = 8$, $|S_3| = 6$).

- identity: $c_V = 8$.
- transpositions (3 elements, order 2, $c(\sigma) = 2$): $c_V = (8+4)/2 = 6$.
- 3-cycles (2 elements, order 3, $c(\sigma) = 1$, so $|V_\sigma| = |V_{\sigma^2}| = 2$): $c_V = (8+2+2)/3 = 4$.

Type	Number	c_V	$\lvert\mathrm{Fix}\rvert = 2^{c_V}$
identity	1	8	256
transposition	3	6	64
3-cycle	2	4	16

Number of classes: $\frac{1}{6}(256 + 3\cdot 64 + 2\cdot 16) = \frac{480}{6} = \boxed{80}$.

$n = 4$ ($|V| = 16$, $|S_4| = 24$).

- identity: $c_V = 16$.
- transpositions (6 elements, order 2, $c(\sigma) = 3$): $c_V = (16+8)/2 = 12$.
- double transpositions (3 elements, order 2, $c(\sigma) = 2$): $c_V = (16+4)/2 = 10$.
- 3-cycles (8 elements, order 3, $c(\sigma) = c(\sigma^2) = 2$): $c_V = (16+4+4)/3 = 8$.
- 4-cycles (6 elements, order 4; $c(\sigma) = c(\sigma^3) = 1$, and σ^2 is a double transposition with $c = 2$): $c_V = (16+2+4+2)/4 = 6$.

Type	Number	c_V	$\lvert\mathrm{Fix}\rvert = 2^{c_V}$
identity	1	16	65536
transposition	6	12	4096
double transposition	3	10	1024
3-cycle	8	8	256
4-cycle	6	6	64

Number of classes:

$$\frac{1}{24}\big(65536+6\cdot4096+3\cdot1024+8\cdot256+6\cdot64\big) = \frac{65536+24576+3072+2048+384}{24} = \frac{95616}{24} = \boxed{3984}.$$

■

Exercise 18. *How many equivalence classes of switching functions are there if the input variables x_1, x_2, x_3, x_4 can be permuted by any permutation in the subgroup of S_4 generated by the permutation $(x_1x_2x_3x_4)$?*

Solution. Let $H = \langle\sigma\rangle = \{\mathrm{e}, \sigma, \sigma^2, \sigma^3\}$ with $\sigma = (x_1x_2x_3x_4)$. As in Exercise 17, a switching function is a 2-coloring of $V = \{0,1\}^4$, and $|\mathrm{Fix}(\tau)| = 2^{c_V(\tau)}$. From Exercise 17:

- $\tau = \mathrm{e}$: $c_V = 16$, so $2^{16} = 65536$ fixed functions.
- $\tau = \sigma, \sigma^3$ (4-cycles): $c_V = 6$, so $2^6 = 64$ each.
- $\tau = \sigma^2 = (x_1x_3)(x_2x_4)$ (double transposition): $c_V = 10$, so $2^{10} = 1024$.

By Burnside's theorem, the number of equivalence classes is

$$\frac{1}{4}\big(65536 + 64 + 1024 + 64\big) = \frac{66688}{4} = \boxed{16672}.$$

■

Exercise 19. *A striped necktie has 12 bands of color. Each band can be colored by one of four possible colors. How many possible different-colored neckties are there?*

Solution. Number the bands $1, \ldots, 12$ from one end to the other, so $S = \{1, \ldots, 12\}$ and $|C| = 4$. There are 4^{12} colorings. A necktie can be turned end for end, which reverses the order of the bands. This gives the group $G = \{\mathrm{e}, \rho\} \cong \mathbb{Z}_2$, where

$$\rho = (1\ 12)(2\ 11)(3\ 10)(4\ 9)(5\ 8)(6\ 7)$$

has six 2-cycles. By (1): $|\mathrm{Fix}(\mathrm{e})| = 4^{12} = 16777216$ and $|\mathrm{Fix}(\rho)| = 4^6 = 4096$. By Burnside's theorem the number of neckties is

$$\frac{1}{2}\big(4^{12} + 4^6\big) = \frac{16777216 + 4096}{2} = \boxed{8390656}.$$

Remark. This treats the tie as reversible (a coloring and its reversal are the same tie). If the two ends were physically distinguishable so that no reversal is allowed, the group is trivial and the answer would be $4^{12} = 16777216$.

■

Exercise 20. *A group acts* faithfully *on a G-set X if the identity is the only element of G that leaves every element of X fixed. Show that G acts faithfully on X if and only if no two distinct elements of G have the same action on each element of X.*

Solution. Say that $g, h \in G$ have the same action if $gx = hx$ for all $x \in X$.

($\Rightarrow$) Suppose G acts faithfully and g, h have the same action. For every $x \in X$ we have $gx = hx$, hence

$$(h^{-1}g)x = h^{-1}(gx) = h^{-1}(hx) = (h^{-1}h)x = \mathrm{e}x = x.$$

So $h^{-1}g$ fixes every element of X, and by faithfulness $h^{-1}g = \mathrm{e}$, i.e. $g = h$. Thus distinct elements never have the same action.

($\Leftarrow$) Suppose no two distinct elements have the same action, and let $g \in G$ fix every $x \in X$. Then $gx = x = \mathrm{e}x$ for all x, so g and e have the same action. By hypothesis $g = \mathrm{e}$. Thus e is the only element fixing every point, so the action is faithful. ■

Exercise 21. *Let p be prime. Show that the number of different abelian groups of order p^n (up to isomorphism) is the same as the number of conjugacy classes in S_n.*

Solution. Recall that a *partition* of n is a nonincreasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$ with $\lambda_1 + \cdots + \lambda_k = n$. We show both quantities equal the number of partitions of n.

Conjugacy classes of S_n. Two permutations in S_n are conjugate iff they have the same cycle type, i.e. the same multiset of cycle lengths (counting 1-cycles). The cycle lengths of a permutation of $\{1, \ldots, n\}$ form a partition of n, and every partition of n occurs (take disjoint cycles of those lengths). So the conjugacy classes of S_n are in bijection with the partitions of n.

Abelian groups of order p^n. By the Fundamental Theorem of Finite Abelian Groups, every abelian group of order p^n is isomorphic to

$$\mathbb{Z}_{p^{\lambda_1}} \times \mathbb{Z}_{p^{\lambda_2}} \times \cdots \times \mathbb{Z}_{p^{\lambda_k}}, \qquad \lambda_1 \geq \cdots \geq \lambda_k \geq 1,$$

where the order is $p^{\lambda_1 + \cdots + \lambda_k} = p^n$ exactly when $\lambda_1 + \cdots + \lambda_k = n$, i.e. (λ_i) is a partition of n. Moreover the theorem says the exponents λ_i (the orders of the cyclic factors) are uniquely determined by the group. Hence $(\lambda_1, \ldots, \lambda_k) \mapsto \mathbb{Z}_{p^{\lambda_1}} \times \cdots \times \mathbb{Z}_{p^{\lambda_k}}$ is a bijection from partitions of n onto isomorphism classes of abelian groups of order p^n.

Composing the two bijections, the number of abelian groups of order p^n up to isomorphism equals the number of conjugacy classes of S_n (both are the number of partitions of n). ■

Exercise 22. *Let $a \in G$. Show that for any $g \in G$, $gC(a)g^{-1} = C(gag^{-1})$.*

Solution. Here $C(a) = \{x \in G : xa = ax\}$.

($\subseteq$) Let $x \in C(a)$. Then

$$(gxg^{-1})(gag^{-1}) = gxag^{-1} = gaxg^{-1} = (gag^{-1})(gxg^{-1}),$$

so $gxg^{-1} \in C(gag^{-1})$.

($\supseteq$) Let $y \in C(gag^{-1})$ and put $x = g^{-1}yg$, so $y = gxg^{-1}$. Since $y(gag^{-1}) = (gag^{-1})y$,

$$xa = g^{-1}yg\,a = g^{-1}\,y\,(gag^{-1})\,g = g^{-1}\,(gag^{-1})\,y\,g = a\,g^{-1}yg = ax.$$

So $x \in C(a)$ and $y = gxg^{-1} \in gC(a)g^{-1}$.

Hence $gC(a)g^{-1} = C(gag^{-1})$. ■

Exercise 23. *Let $|G| = p^n$ and suppose that $|Z(G)| = p^{n-1}$ for p prime. Prove that G is abelian.*

Solution. Since $Z(G)$ is a normal subgroup, the quotient $G/Z(G)$ is a group of order $p^n/p^{n-1} = p$. A group of prime order is cyclic, so $G/Z(G) = \langle gZ(G) \rangle$ for some $g \in G$.

We show that a group G with $G/Z(G)$ cyclic is abelian. Every $x \in G$ lies in some coset $g^i Z(G)$, so $x = g^i z$ with $z \in Z(G)$. Let $x = g^i z$ and $y = g^j w$ with $z, w \in Z(G)$. Because z, w are central,

$$xy = g^i z g^j w = g^i g^j z w = g^{i+j} z w, \qquad yx = g^j w g^i z = g^{j+i} w z = g^{i+j} z w.$$

So $xy = yx$ for all $x, y \in G$, and G is abelian. (In fact this forces $Z(G) = G$, so $|Z(G)| = p^{n-1}$ is impossible for a nonabelian group of order p^n.) ■

Exercise 24. *Let G be a group of order p^n where p is prime and X a finite G-set. If $X_G = \{x \in X : gx = x \text{ for all } g \in G\}$, prove that $|X| \equiv |X_G| \pmod{p}$.*

Solution. The orbits of G partition X. For $x \in X$, Theorem 1 gives $|\mathcal{O}_x| = |G|/|G_x| = p^n/|G_x|$, and $|G_x|$ divides $|G| = p^n$ (Lagrange), so $|\mathcal{O}_x|$ is a power of p. Moreover

$$|\mathcal{O}_x| = 1 \iff gx = x \text{ for all } g \in G \iff x \in X_G.$$

Hence the orbits of size 1 are exactly the singletons $\{x\}$ with $x \in X_G$, and every other orbit has size p^m with $m \geq 1$, which is divisible by p. Let $\mathcal{O}_{x_1}, \ldots, \mathcal{O}_{x_r}$ be the orbits of size greater than 1. Then

$$|X| = |X_G| + \sum_{i=1}^{r} |\mathcal{O}_{x_i}| \equiv |X_G| \pmod{p}.$$

■

Exercise 25. *If G is a group of order p^n, where p is prime and $n \geq 2$, show that G must have a proper subgroup of order p. If $n \geq 3$, is it true that G will have a proper subgroup of order p^2?*

Solution. **Step 1: $Z(G)$ is nontrivial.** Let G act on $X = G$ by conjugation, $g \cdot x = gxg^{-1}$. Then

$$X_G = \{x : gxg^{-1} = x \ \forall g\} = Z(G).$$

By Exercise 24, $|Z(G)| \equiv |G| = p^n \equiv 0 \pmod{p}$ (for $n \geq 1$). Since $\mathrm{e} \in Z(G)$, $|Z(G)| \geq 1$, so $|Z(G)| \geq p$. In particular, $Z(G)$ contains an element $z \neq \mathrm{e}$.

Step 2: a subgroup of order p. The order of z divides p^n, so $\mathrm{ord}(z) = p^k$ with $k \geq 1$. Then $w = z^{p^{k-1}}$ has order exactly p, and $N = \langle w \rangle$ is a subgroup of order p. Since $w \in Z(G)$, N is a *normal* subgroup of G. Because $n \geq 2$, $|N| = p < p^n = |G|$, so N is proper. This proves the first claim.

Step 3: a subgroup of order p^2 when $n \geq 3$. The answer is **yes**. We prove more generally:

Lemma 4. *If $|G| = p^n$, then G has a subgroup of order p^k for every $0 \leq k \leq n$.*

Proof. Induct on n. The cases $k = 0$ (trivial subgroup) and $n = 0, 1$ are clear, so let $n \geq 2$ and assume the lemma for all groups of order p^m with $m < n$. Let $1 \leq k \leq n$. By Step 2, G has a normal subgroup N of order p. The quotient G/N has order p^{n-1}, so by the induction hypothesis it has a subgroup $\overline{H}$ of order p^{k-1} (note $0 \leq k-1 \leq n-1$). Let $\pi\colon G \to G/N$ be the canonical homomorphism and $H = \pi^{-1}(\overline{H})$. Then H is a subgroup of G containing $N = \ker \pi$, and $\pi|_H\colon H \to \overline{H}$ is onto with kernel N, so

$$|H| = |N| \cdot |\overline{H}| = p \cdot p^{k-1} = p^k.$$ □

Taking $k = 2$ in the lemma, a group of order p^n with $n \geq 3$ has a subgroup of order p^2, and this subgroup is proper because $p^2 < p^n$. ■